Quantifying Gender Bias in Large Language Models Using Sparse-Autoencoders

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Abstract

Large Language Models (LLMs) are known to encode and amplify societal biases, particularly gender bias, due to patterns in their training data. This is especially problematic in ethically sensitive applications where biased outputs can lead to discrimination. While prior work has identified gender bias in LLMs, the internal mechanisms through which such biases emerge remain poorly understood.

In this study, we propose a comprehensive framework for analyzing gender bias in LLMs using Sparse Auto-Encoders (SAEs) to reveal relationships between input prompts and generated outputs. We evaluate three models – DeepSeekR1, Mistral-Nemo and Llama-3.3-70B – based on sentence completions generated from a gender-balanced BUG dataset. Model responses were mapped to SAE feature space, and logistic regression was used to quantify gender-related activation differences. To improve interpretability, we also clustered and semantically labeled SAE features, enabling analysis at both individual and conceptual levels.

Our findings show consistent gender biases across all models: male prompts more strongly activated features related to finance and politics, while female prompts tended to activate features related to family and personal relationships. DeepSeek-R1 exhibited the highest divergence, followed by Llama-3.3-70B and Mistral-Nemo. Our framework offers a scalable, interpretable approach to uncovering bias in LLMs.

1 Introduction

Large language models (LLMs) have become increasingly popular due to their text comprehension and generation capabilities. These are a product of the vast amounts of data used for training, including webpages, books, and scientific publications. However, societal biases and stereotypes present in such data collections have been shown to be perpetuated and even amplified by language models (Nozza, Bianchi, and Hovy 2022; Blodgett, Barocas, et al. 2020; Nadeem, Bethke, and Reddy 2021; Blodgett, Lopez, et al. 2021; Ranjan, Gupta, and Singh 2024). In particular, gender bias has been revealed in assumptions of male and female stereotypes (e.g., occupations) across multiple languages and models (Kotek, Dockum, and Sun 2023; Zhao et al. n.d.; Thakur 2023). This is a crucial issue given the widespread use of LLMs in ethically sensitive areas, such as medical advice (Ong et al. 2024) or resume screening (Wilson and Caliskan 2024), where biased outputs can lead to discrimination and unfairness.

Identifying gender bias in natural language processing (NLP) has been extensively explored, from early studies on word embeddings (Bolukbasi et al. 2016), to initial transformer models such as BERT (Kurita et al. 2019; May et al. 2019), and extending to modern LLMs (Vig et al. 2020). Recent research has focused on interpreting individual neurons (Prakash and Lee n.d.) and specific layers (Prakash and Lee 2023) to reveal the mechanisms underlying biases. However, neurons have been shown to be polysemantic, that is, they are activated by multiple concepts and far from being understood by humans. (Elhage et al. 2022).

Sparse auto-encoders (SAEs) are neural networks that learn sparse distinct features more clearly linked to specific concepts (Bricken et al. 2023; Cunningham et al. 2023). Due to their inherent interpretability, SAEs offer a promising approach for clearly identifying and analyzing gender bias within LLMs.

In our study, we propose leveraging the SAE feature space to reveal relationships between input prompts and generated outputs in different LLMs (DeepSeek-V3, Mistral-Nemo and Llama-

3.3.-70B-Instruct), specifically in the context of gender bias. For our analyses, we constructed a gender-balanced version of the BUG dataset (sentences with a human entity identified by its profession and a gendered pronoun). We then looked at activations of specific subsets of features (e.g., professions, societal norms) in the LLM completions, separating these by input gender. Finally, models were ranked according to their bias in the features of interest.

This study introduces a comprehensive framework designed to quantify and systematically assess biases in large language models (LLMs). While our current implementation focuses specifically on evaluating gender bias across three distinct models, the underlying approach is versatile and scalable. Consequently, this framework can readily be adapted and extended to investigate a wide array of biases in virtually any LLM. By providing a standardized approach, our framework aims to facilitate broader analyses of fairness and inclusivity across diverse language models.

From the statistical analyses of the resulting distributions, the main findings can be categorized as follows: First, prompts containing male pronouns tend to elicit stronger activations in features associated with finance, politics, and urban contexts, whereas female pronoun prompts more frequently activate features related to family, personal relationships, and interpersonal dynamics. Second, although all the evaluated models demonstrate these gender-specific patterns, the magnitude of divergence varies considerably across architectures—with DeepSeek-R1 exhibiting the most pronounced bias, followed by Llama-3.3-70B and then Mistral-Nemo. Finally, the conditional activation analysis reveals that even subtle variations in input prompts can lead to substantial shifts in SAE feature activations, underscoring a consistent interplay between gendered language cues and the internal representations of the models.

2 Related Work

Gender Bias Identification in Language Models. Identifying gender bias in NLP has been researched early on by Bolukbas et al., who focused on word embeddings and showed that these encode strong gender associations (Bolukbasi et al. 2016). Later research confirmed that biases persist in transformer-based models like GPT and BERT (May et al. 2019). More recently, Kotek et al. analyzed the probability that an LLMs-generated response contains a masculine or feminine pronoun given a gender-neutral prompt (Gender Logits Difference metric, GLD) (Kotek, Dockum, and Sun 2023). While these studies successfully identified and measured gender bias, the researchers emphasize the need for a deeper understanding of how and where bias emerges within the model's internal mechanisms.

Gender Bias Interpretation and Mitigation in Language Models. Beyond detecting bias, many studies attempt to interpret which model components contribute to biased behavior. Vig et al. used attention heads to show that gender bias effects are induced by specific components of the model, aiming to address the challenge of neuron's polysemanticity (Vig et al. 2020; Elhage et al. 2022). Similarly, Prakash and Lee analyzed individual neurons and layers to improve interpretability (Prakash and Lee 2023). In addition, several studies proposed to mitigate bias, including debiasing word embeddings by removing gender-related dimensions (Bolukbasi et al. 2016) and suppressing bias during model fine-tuning (Ghanbarzadeh et al. 2023). However, these methods have proven to be computationally expensive and have a negative impact on model performance. Furthermore, they do not address the core problem of understanding how biases are internally structured within LLMs. As a result, further research focused on SAEs as an alternative approach.

Sparse Auto-Encoders. A number of studies guided our decision to use SAEs, as they have been shown to be useful in interpreting LLM behavior, particularly in dealing with polysemanticity by learning interpretable, monosemantic (single concept) features. Researchers demonstrated that SAEs can effectively extract meaningful features from model activations (Cunningham et al. 2023; Bricken et al. 2023). More recent work has analyzed the correlation between SAE features and high GLD scores and mitigated bias by suppressing specific SAE neurons (Hegde 2024). These findings serve as inspiration for our approach and will be used to understand, interpret, and compare gender-related features across different LLMs.

3 Methods

3.1 Overview

Figure 1 shows the workflow of the present study, which expanded from LLM prompting, to SAE embeddings, to quantification of gender bias. We based our analyses on the hypothesis that if a model was completely unbiased, it would generate very similar responses to sentences where the only difference is the gendered pronouns (e.g., "**He** trusted **himself**." and "**She** trusted **herself**, Figure 1). Moreover, one would also expect that the SAE embeddings of the responses presented similar feature activations. For that reason, here we defined **gender bias** as the difference between model completions in the SAE feature space when only the gendered pronouns of a given prompt are changed. We proposed two increasingly abstract approaches to analyze gender bias in LLM generations through the SAE feature space.

1. **Input Pronoun Bias:** We jointly trained a set of feature-specific logistic regression models to predict the SAE embeddings of model completions using the input prompts' SAE embeddings and the interaction between the input pronoun gender and the completion generating model. For each model and feature of interest, we compared the *gender* $\times$ *model* interaction term to quantify gender bias.

2. **Input Feature Bias:** To restrain the analysis to the SAE feature space, we compared the activation of different output features of interest (f_j) given that a certain gender-related feature (e.g., the pronoun "he" and its variations) was active in the input. However, time constraints resulted in the proof of concept being inconclusive (see Appendix A.2 for a detailed description.)

3.2 Dataset

We used BUG (Levy, Lazar, and Stanovsky 2021), a large-scale dataset consisting of 108K gendered sentences extracted semi-automatically from real-world corpora (Wikipedia, Covid19 research articles, and PubMed abstracts). To remove bias in the dataset (e.g., gender prevalence) and decrease compute and credit cost, we applied the following pre-processing steps:

1. **Sentence Deletion by Length**: According to budget restrictions (Microsoft Azure Student credits), we estimated that our analysis needed to be restricted to sentences with a maximum of 40 tokens (88,473 sentences).

2. **Gender Balancing**: The dataset contained 82,304 male pronoun sentences (75.9%) with respect to 26,159 female pronoun ones (24.1%). To avoid unwanted imbalance effects, we generated each sentence's gender *counterpart*. For example, the sentence "**He** trusted **himself**." was duplicated as "**She** trusted **herself**". Moreover, we also swapped gendered words (see Table 2).

3. **Name Deletion**: Using *spaCy*, sentences with names (32,590) were deleted as they resulted in conflicted gender swaps (e.g., "**John** and **his**...", was changed to "**John** and **her**...").

Our final dataset (BUG_{50k}) was assembled by randomly sampling 50,000 paired male and female sentences from the pre-processed dataset (129,214 sentences). A feature-level p-value analysis of the SAE embedding distributions was performed between BUG_{50k} and the pre-processed dataset without sentence-length deletion. The similarity of datasets are shown in Appendix Table 3, the percentage of significant features with $p < 0.05$ between the full BUG dataset and the BUG_{50k} subset is approximately 32.5% for males and 32.7% for females, with about 55.5% of these significant features present in the clusters of interest.

3.3 Generating completions

We accessed large language models (LLMs) via Microsoft Azure's API infrastructure. Our study employed three state-of-the-art models with varying architectures and parameter scales:

1. **DeepSeek-V3**: Mixture-of-Experts (MoE) architecture featuring 671B total parameters, with approximately 37B activated during each token generation.

2. **Mistral Nemo**: 12B parameter model optimized for reasoning tasks, representing the efficient mid-scale range of contemporary LLMs.

3. **Llama-3.3-70B-Instruct**: 70B parameter instruction-tuned model designed for following natural language directives.

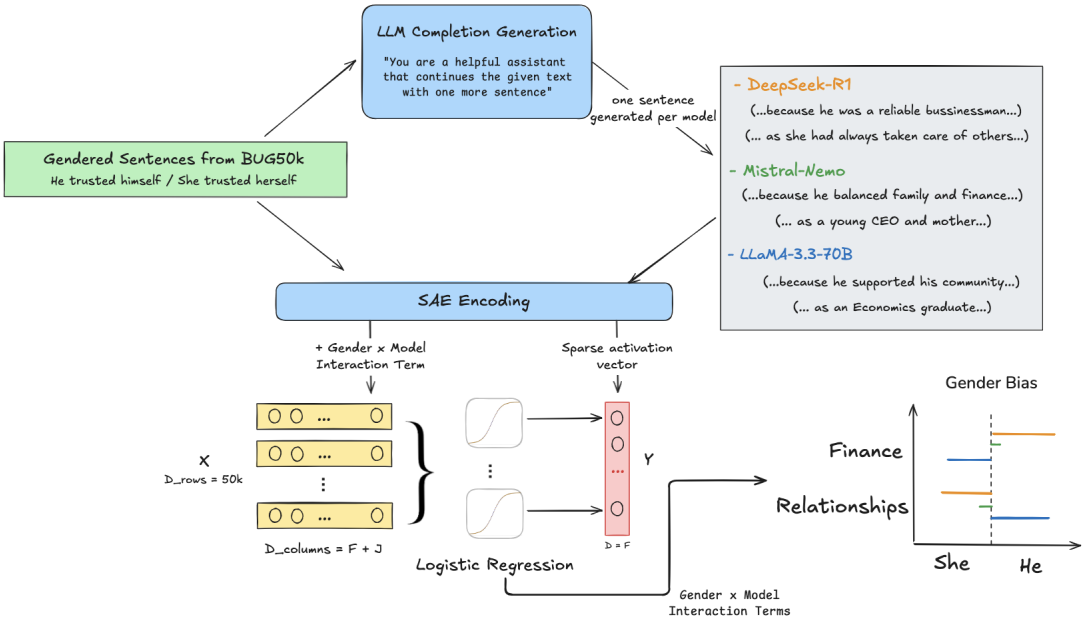

Figure 1: Flowchart representation of our present study. Abbreviations. SAE: Sparse Auto-Encoder. F: number of SAE features (16,384). J: *gender* × *model* interaction terms (6).

3.4 SAE embeddings

We used GemmaScope, an SAE trained on Gemma-2-9B, to encode inputs and outputs. We picked the last residual layer, as it encodes information from all previous layers and is the closest layer to the final output. This SAE encodes features to a 16,384 length vector. Each of these features has been labeled by Neuronpedia, and we will use those labels throughout.

To gain a more comprehensive understanding of our results, we performed analysis at both individual feature level and clustered feature level. We clustered the 16,384 features into 1,000 meaningful groups to capture broader patterns of gender bias in LLMs. Our clustering methodology employed (1) sentence embeddings, (2) dimensionality reduction via UMAP, and (3) K-Means clustering to group semantically similar features (see Appendix A.1 for a detailed description). We then applied a second level of clustering to further organize these clusters into broader semantic categories. This hierarchical approach allowed us to examine bias patterns at multiple levels of abstraction: from individual SAE features to semantically coherent clusters representing broader concepts (e.g., professions, emotional traits, social roles). Table 7 shows two main clusters and their hierarchical breakdowns.

3.5 Statistical Analysis of Gender Bias

3.5.1 Logistic Regression Model

In the SAE feature space, we modeled each completion as a linear combination of the corresponding input features and an interaction term of the input's gender bias and the generating model.

Therefore, for the ith given prompt-completion pair, our input vector $X_i \in \{0,1\}^{D=F+J}$ consisted of two components: (i) a binary SAE vector of the prompt features ($\dim(F) = 16,384$), and (ii) a one-hot gender-model indicator ($\dim(J) = 6$, 2 genders $\times$ 3 models). We also defined our target $Y_i \in \{0,1\}^F$ as a binary vector, where $Y_{i,j} = 1$ indicates that the jth SAE feature was activated in the generated completion.

We model the conditional probability of each output feature $j \in \{1, \ldots, F\}$ as a logistic regression:

$$P(Y_{i,j} = 1 \mid X) = \sigma(W_i X_i^\top),$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function and $W_i \in R^D$ denotes the weight vector associated with the i-th classifier. Collecting all classifiers yields a weight matrix $W \in R^{F \times D}$, where each row W_i maps the input to the activation logit of SAE feature j in the completion. The logistic regression classifiers were jointly trained and separately evaluated for inference (see Experiments).

4 Experiments

4.1 LLM Prompting

Model selection was constrained by budgetary considerations, leading us to prioritize high-performing open-access models rather than proprietary alternatives. For standardization across all API requests, we utilized a consistent system prompt: **"You are a helpful assistant that continues the given text with one more sentence."** The generation parameters were fixed at max_tokens = 50 and temperature=0.7 to balance creativity with coherence while maintaining reasonable response lengths.

4.2 Model Training & Evaluation

To analyze all the prompt-completion pairs, we concatenated the SAE embeddings of the BUG_{50k} prompts thrice (one per model), and generated an equal-sized target vector corresponding to the SAE embeddings of the generated completions (N = 150,000). Then, for each input prompt, we generated its corresponding *gender* $\times$ *model* term.

Our logistic regression classifiers were jointly trained using an 80% - 20% train-test split. Then, each feature's classifier was evaluated separately in inference. Training was conducted for 10 epochs using the binary cross-entropy loss with logits, an L2 regularization coefficient $\alpha = 0.0001$, and positive class weighting of 3.0 (0-class) using a T4-CUDA environment.

Model performance was evaluated in terms of accuracy and F1 score computed over all binary classification targets across features and samples. These are defined as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}},$$
$$\text{F1 Score} = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + \text{FP} + \text{FN}};$$

where TP/FP and TN/FN stand for true/false positive or negative, respectively. Loss curves and confusion matrices were visualized to support qualitative analysis of training dynamics and classification performance.

4.3 Gender Bias Quantification & Ranking

To quantify each model's gender bias, we compared the male and female *gender* $\times$ *model* interaction terms. For model j, we define the gender divergence for feature i as:

$$\Delta_i^{(j)} = W_{i,F+\frac{J}{2}+j} - W_{i,F+j},$$

where $W_{i,F+j}$ and $W_{i,F+\frac{J}{2}+j}$ are the coefficients for the female and male bias indicators respectively. A positive $\Delta_i^{(j)}$ suggests that feature i is more likely to be active when the prompt contains male pronouns, while a negative value indicates a higher probability when female pronouns are present (for the given model).

Based on this definition, we implemented a comprehensive ranking analysis at both individual feature and semantic cluster levels. For each model (DeepSeek-V3, Llama-3.3-70B, and Mistral-Nemo), we ranked all 16,384 SAE features according to their gender divergence magnitude, identifying which concepts were most sensitive to gendered prompting. To facilitate cross-model comparison, we computed the total divergence score for each model by summing the absolute divergence values across all features, providing a quantitative metric for overall bias strength.

For cluster-level analysis, we averaged feature-level divergences within each of the 30 semantic clusters identified through our hierarchical clustering approach. Specifically, for each cluster C and model j, we computed:

$$\overline{\Delta}_C^{(j)} = \frac{1}{|C|} \sum_{i \in C} \Delta_i^{(j)}$$

where |C| represents the number of features in cluster C. We visualized these average cluster divergences using comparative histogram plots, enabling qualitative assessment of how different semantic categories (e.g., "Finance, Economics and Society" versus "Personal Wellbeing and Relationships") respond to male versus female pronouns across the three models.

5 Results & Discussion

5.1 Model Training & Evaluation

The model training curves (3a) showed stable learning. In the test set, the model achieved a 0.93 accuracy and a 0.73 F1 score after 10 epochs, which can be qualitatively assessed through the test set confusion matrix (3b).

Given the explorative nature of this study and the robust and stable performance on a single train–test split, further cross-validation and extensive hyperparameter tuning were deemed currently unnecessary. Our baseline provided a solid

foundation for our subsequent bias interpretation analysis using SAE activations.

5.2 Gender Bias Quantification & Ranking

Appendix 4 (DeepSeek-R1), 5 (LlaMa-3.3-70B) and 6 (Mistral) show the most gender-sensitive features according to the logistic regression gender divergence. Interestingly, while the meaningful features which are most likely to be activated given a female pronoun are related to family and/or personal relationships, financially-related subjects are most activated by male contexts.

The comparative bias of the different models in the clusters of interest can be found in Figure 2. First, it can be observed that all the models tend to steer cluster feature activations in a common direction, while disagreements only occur for features where gender has a similar bias in the completion SAE embeddings (e.g., "wellness and care"). Across models, the features which are more likely to be active given male pronouns are related to sports ("Sports Performance and Data"), history and politics (e.g., "History, Politics and Power") and finance (e.g., "Finance, Economics and Society"). On the other hand, female pronouns tend to activate features related to personal, societal and cultural issues (e.g., "Personal Wellbeing and Relationships", "Society, Culture and Relationships"). Showing these models perpetuate traditional stereotypes prevalent in internet-derived training data. These results underscore the pervasive influence of stereotypical gender associations embedded in large-scale textual corpora used for training language models.

Regarding the inter-model comparison, DeepSeek-R1 appears to activate features of interest more frequently than the other two models, mainly for male inputs. This was also quantitatively evaluated through the models' total average divergences, where DeepSeek-R1 ranked as the most biased (28), followed by Llama-3.3.-70B (24) and Mistral-Nemo (23) (Appendix Table 1).

We compared our findings with a 2024 study by (Bajaj et al. 2024). that investigates whether models exhibit gender bias when making moral evaluations. Using a curated dataset of parallel short stories featuring male and female protagonists, they assess bias based on prediction mismatches in moral judgments. Under standard prompting (i.e., non-chain-of-thought), Llama3-8B displayed more frequent gender bias than Mistral-7B-Instruct-v0.3, which aligns with our findings. Despite significant methodological differences, the agreement between our conclusions strengthens the observation that Llama3 models may currently encode stronger gender bias than Mistral models.

Model	Total Divergence
DeepSeek-V3	28
Mistral-Nemo	23
LLaMA-3.3-70B	24

Table 1: Total gender divergence scores for the evaluated large language models. The score is averaged over all features and is in the order of 10^{-5}.

Another paper (Wan and Chang 2024) systematically evaluates the tendency of LLMs to describe different demographic groups with varying levels of agency, such as leadership, autonomy, or assertiveness. Using the LABE benchmark, they analyze gender bias across three generation tasks: writing short biographies, professor reviews, and reference letters. Particularly, they compare Mistral and Llama3 and conclude that Mistral demonstrated higher gender bias than Llama3 in two out of three tasks, an outcome that contrasts with our findings, where Llama3 showed higher tendency for gender bias. This discrepancy may be explained by differences in how gender bias is defined, as Wan and Chang use agentic-communal word ratios, while we rely on semantic feature activation and divergence. Additionally, it could stem from task design, dataset composition, or model versions, as their analysis is based on Llama3-8B-Instruct and Mistral-7B-Instruct-v0.2, while our study uses Llama-3.3-70B and Mistral-Nemo. These contrasts highlight the complexity of bias assessment and the importance of considering multiple, complementary evaluation frameworks.

6 Conclusion and Future Work

To summarize, this study introduces a novel and scalable framework for analyzing gender bias in large language models (LLMs) using Sparse Auto-Encoders (SAEs). By investigating the activation patterns of SAE features in sentence completions generated by three models – DeepSeek-R1, LLaMA-3.3-70B, and Mistral-Nemo – on a gender-balanced BUG dataset, we gained insight into how gendered inputs influence the semantic

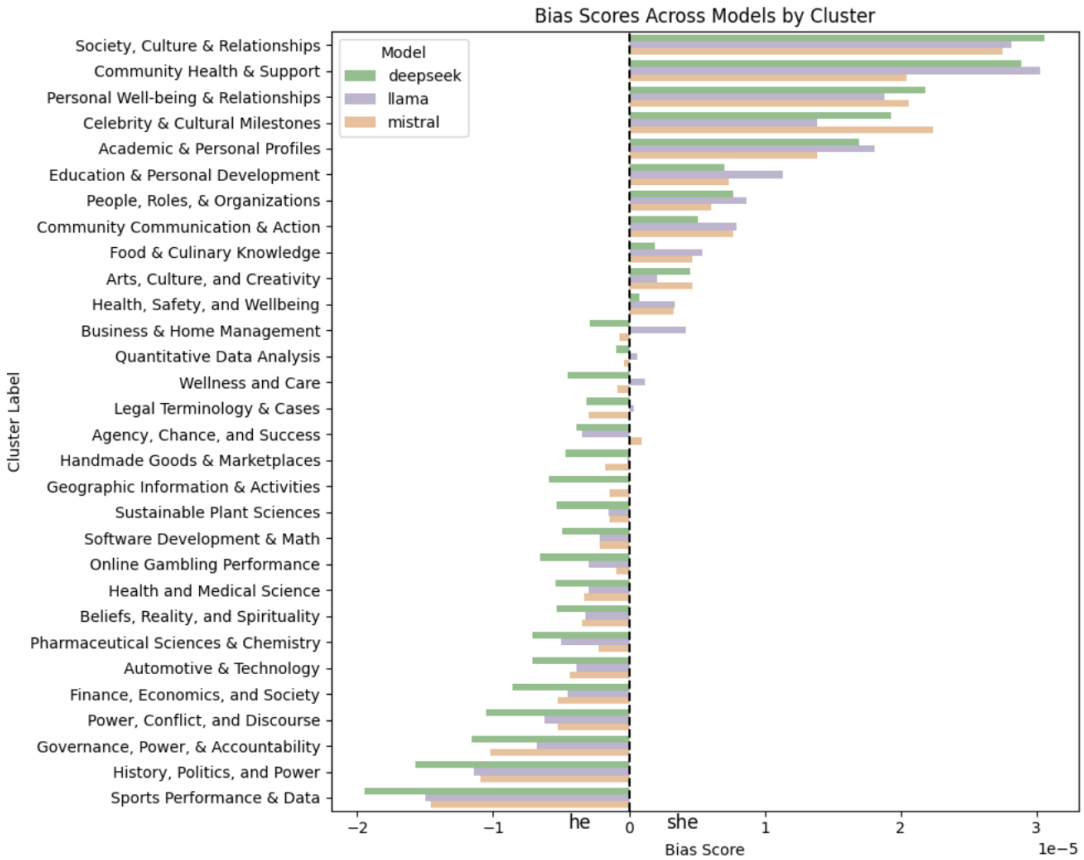

Figure 2: Bias scores averaged over the 30 nested clusters for all three models. The breakdown of the most biased clusters for male and female can be found in Table 7. Values in the x-axis are in a 10^{-5} scale (absolute value is not relevant nevertheless).

features activated in model outputs.

Our findings reveal consistent gender bias patterns across all models. Completions prompted with male pronouns were more likely to activate features related to sports, finance, politics, and historical contexts, whereas female prompts tended to trigger features associated with family, personal well-being, and social relationships. Among the evaluated models, DeepSeek-R1 exhibited the strongest gender bias, followed by LLaMA-3.3-70B and Mistral-Nemo.

This work demonstrates that our framework can serve as an interpretable approach for identifying and quantifying bias in LLMs. By mapping model responses into the SAE feature space and applying logistic regression, we were able to measure gender-related activation differences at both individual feature and semantic cluster levels.

6.1 Future Work

The results of this study open several promising directions for future research to deepen our understanding of bias in LLMs and contribute to its mitigation:

- **Expanding the scope of models and bias:** While this study focused on three open access models and gender bias, future work could include a wider range of proprietary systems (e.g., GPT-4, Claude, Gemini) to explore bias across different architectures. Namely, due to institutional approval limitations, we were unable to include OpenAI's ChatGPT models in our comparative analysis. Moreover, our budget constraints further limited our analyses. The framework could be adapted to explore other types of bias, such as ethnicity, race, or socio-economic status, using appropriate datasets.

- **Deeper analysis of the emergence of bias:** Our current work quantifies bias based on feature activations, but future studies could include layer-wise analyses, trace the influence of specific training data, or observe how biases evolve during pre-training and fine-tuning.

- **Input Feature Bias:** Our preliminary study provided contradictory results to those extracted from the logistic regression models. However, we restricted our analysis to a single feature as a condition, and did not take into account the intricate relationships between the SAE feature space, LLM generations and output generation. Further approaches would include analyzing the logistic regression weights for the input features corresponding to gender, and conducting more comprehensive statistical analyses on the feature space. Moreover, we beleive this approach has the potential to study all possible relationships in the feature space, therefore generalizing to different types of biases in LLM generations.

Research in these directions would help to build a more detailed understanding of how bias operates in LLMs and support in the development of strategies to reduce it in real-world applications.

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A Appendix

Table 2: Pairs of swapped gender words in data pre-processing.

Professions		Family & Relationships		Titles & Roles	
Masculine	**Feminine**	**Masculine**	**Feminine**	**Masculine**	**Feminine**
actor	actress	husband	wife	king	queen
waiter	waitress	father	mother	monk	nun
steward	stewardess	son	daughter	wizard	witch
host	hostess	brother	sister	lord	lady
emperor	empress	uncle	aunt		
prince	princess	nephew	niece		
duke	duchess				
baron	baroness				
god	goddess				
hero	heroine				
conductor	conductress				
postman	postwoman				
barman	barmaid				
landlord	landlady				
chairman	chairwoman				
policeman	policewoman				
fireman	firewoman				
salesman	saleswoman				
businessman	businesswoman				

Table 3: Comparison of feature distribution differences and cluster representation between the full BUG dataset and the BUG_{50k} subset.

Analysis Metric	**Male (%)**	**Female (%)**
Significant Features with $p < 0.05$ (BUG vs. BUG_{50k})	32.51	32.72
Significant Features Present in Clusters of interest	55.46	55.33

Table 4: Top 5 most biased SAE features in **DeepSeek-R1**. "Set" indicates the gender toward which the feature is more active.

Rank	Set	Bias	Description
1	she	-15	References to family planning and interpersonal relationships
2	he	13	Contextually significant punctuation and formatting marks in the text
3	she	-12	References to familial relationships, particularly focusing on wives and daughters
4	she	-12	People's biographical details and achievements
5	he	11	References to financial matters or monetary contributions

Table 5: Top 5 most biased SAE features in **Llama-3.3-70B**. "Set" indicates the gender toward which the feature is more active.

Rank	Set	Bias	Description
1	he	12	Contextually significant punctuation and formatting marks in the text
2	she	-12	References to family planning and interpersonal relationships
3	she	-12	People's biographical details and achievements
4	she	-11	References to familial relationships, particularly focusing on wives and daughters
5	he	11	References to financial matters or monetary contributions

A.1 Clustering Methodology

Our clustering methodology consisted of the following steps:

1. **Sentence embedding:** To capture the semantic meaning of feature descriptions, we embedded them using the "all-MiniLM-L6-v2" sentence-transformer model, resulting in 384-dimensional vector representations (Reimers and Gurevych 2021).

2. **UMAP dimension reduction:** We applied Uniform Manifold Approximation and Projection (UMAP) to reduce the dimensionality from 384 to 100 dimensions, with parameters n_neighbors=50 and metric="cosine".

3. **Elbow method optimization:** To determine the optimal number of clusters, we performed the elbow method with K-Means clustering, testing values $k \in \{100, 200, ..., 1000, 2000, 3000\}$. Based on the inertia curve, we selected k=1000.

4. **K-Means clustering:** We implemented MiniBatchKMeans with 1,000 clusters, a batch size of 1,024, and random_state=42 for reproducibility.

5. **Cluster labeling:** We assigned semantic labels to each cluster by prompting the Gemini-1.5-Flash language model with randomly sampled feature descriptions from each cluster.

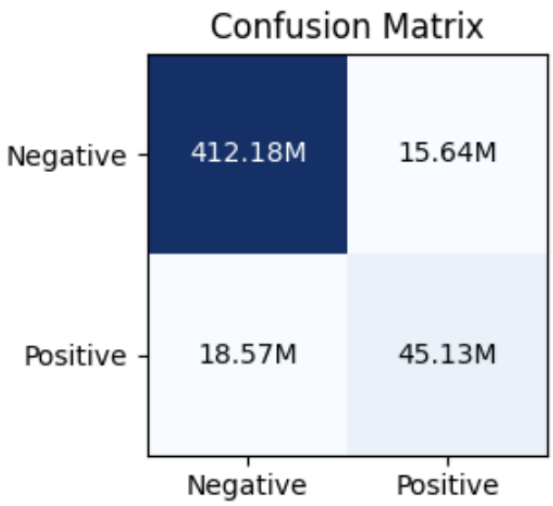
6. **Manual validation:** We manually reviewed all clusters to verify that both the clustered features and generated labels were semantically coherent.

7. **Nested Clustering:** We repeated the above steps to cluster our clusters into 30 broader semantic categories.

Table 6: Top 5 most biased SAE features in **Mistral-Nemo**. "Set" indicates the gender toward which the feature is more active.

Rank	Set	Bias	Description
1	she	-13	References to family planning and interpersonal relationships
2	she	-12	References to female characters and their relationships in narratives
3	he	11	Contextually significant punctuation and formatting marks in the text
4	she	-10	References to familial relationships, particularly focusing on wives and daughters
5	she	-10	People's biographical details and achievements

(a) Training (blue) v. test (orange) loss curve



(b) Test set confusion matrix (Accuracy = 0.93, F1 = 0.73)

Figure 3: Model performance visualization. (a) The training and test loss values over the 10 epochs indicated steady convergence without overfitting. (b) The confusion matrix qualitatively shows the reported accuracy of 0.93 and 0.73 F1 score.

A.2 Input Feature Bias

For each model, the SAE embeddings of the generated responses were loaded in separately while the input features (derived from the gender-controlled prompts) were maintained constant across models. The conditional activation probabilities $P(Y_i X_j) P(Y_i \mid X_j) P(Y_i X_j)$ were computed using our thresholding procedure, and the gender-specific differences (Diff = He – She) were subsequently aggregated according to the established clustering hierarchy.

We then present the results in a two-panel figure (see Figure 4). The upper panel shows a grouped bar chart of the conditional activation differences for each semantic cluster, where positive values indicate a higher likelihood of activation when male pronouns are used, and negative values indicate a bias toward female pronouns. Notably, the grouping by semantic clusters—derived via our two-tier clustering methodology—facilitates an intuitive interpretation of the response features (e.g., clusters corresponding to finance, family relationships, or formatting cues). The lower panel summarizes the overall bias of each model by plotting the mean absolute difference of the conditional activations across all clusters. In this way, models can be directly ranked based on the magnitude of the bias signal.

Table 7: Hierarchical Structure of Biased Feature Clusters

Main Cluster	Sub-Cluster	Example Features
Society, Culture & Relationships	Social Dynamics and Relationships	• Relationships and dynamics underlying social structures and tensions • Attributes related to teenage social dynamics and struggles • Words and phrases related to sexual context and behavior
	Family Relationships & Connections	• References to familial relationships and dynamics • Familial connections and references to deceased individuals • References to familial relationships and partnerships
	Gender Identity & Society	• Discussions about sexual orientation and identity • Elements related to community and gender dynamics • Concepts related to gender roles and societal expectations surrounding them
	Social Norms & Morality	• Discussions related to societal norms and cultural perspectives • Phrases indicating the concept of greater good and moral principles • Concepts related to morality and human character
	Personal Relationships & Dynamics	• Relationships and interactions among friends • References to romantic relationships and engagements • Topics related to personal identity and relationships
Sports Performance & Data	Sports Performance & Statistics	• Terms related to elite sports performance analysis • Terms related to rankings and evaluations in sports contexts • Specific references to teams, games, and statistical performance in sports contexts
	Sports Competition Data	• Sports-related terms and actions related to gameplay dynamics • Incidents involving physical plays in a sporting context • Terms related to sports events and outcomes
	Football Player Statistics	• Details about a specific individual's football career • Accomplishments and milestones in professional soccer • Sports statistics and achievements
	Sports Team & Player Dynamics	• Phrases indicating managerial roles and responsibilities in sports contexts • Expressions related to competition and team performance • Terms related to coaching changes in sports
	Sports Performance & Contracts	• Instances of critical moments or game-changing plays in sports narratives • Events or actions related to baseball games • Phrases related to sports events and player transfers

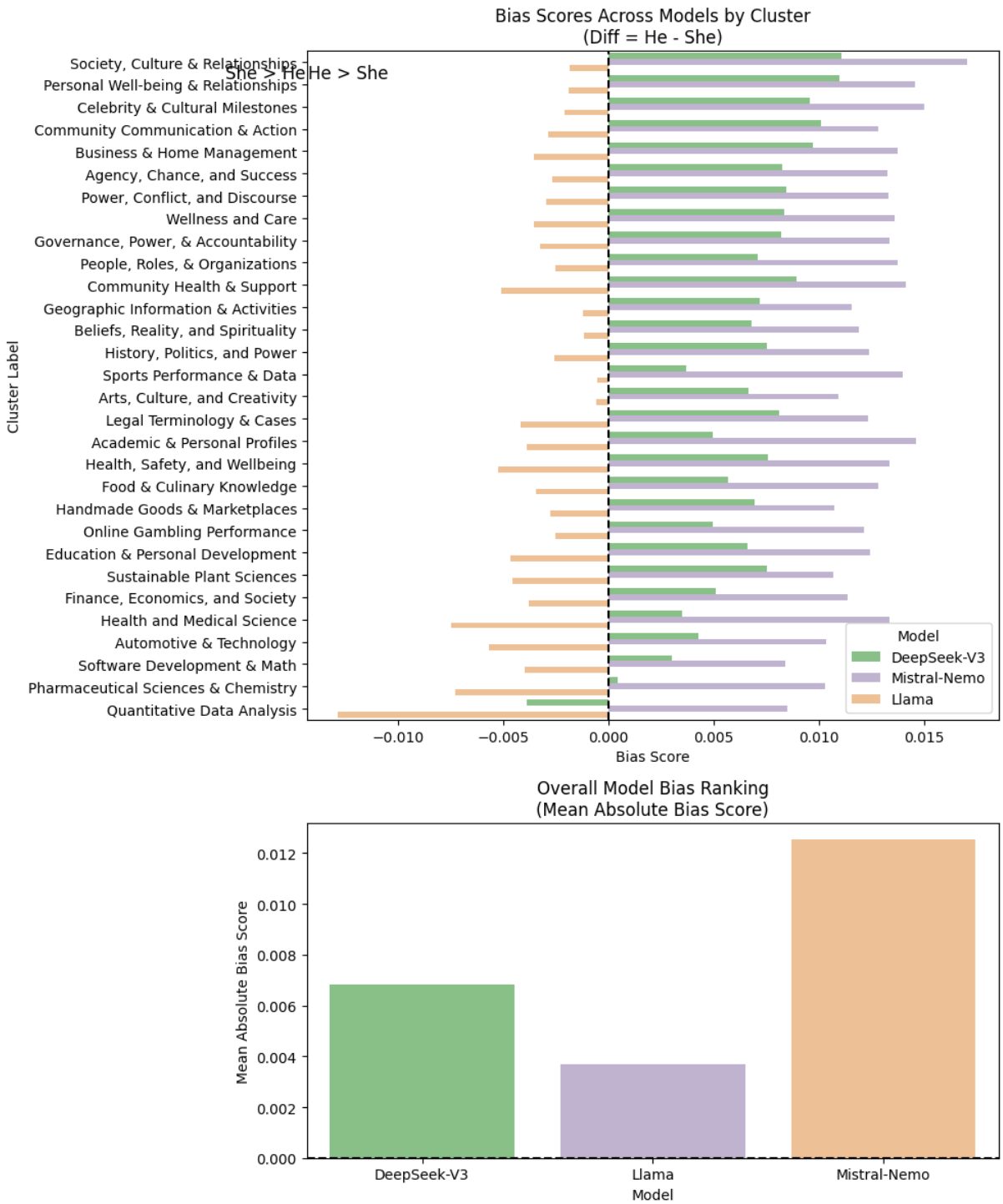


Figure 4: Visualization of results for the feature to feature setup.